

Assignment 6C. Gaussian quadrature

Marks 20

Posted on 17.09.2026 @ 2:30 pm and due on 17.09.2026 @ 5:40 pm

1. Write library functions for Gauss-Legendre and Gauss-Laguerre quadrature for numerical integration. Hardwire the table for zeros and weights for Legendre and Laguerre to any order of your choice (at least $n = 5$). There should be only one Gauss Quadrature routine and depending on what you want it should choose appropriate table for Legendre and Laguerre. [5]
2. Use appropriate $n = 4$ -point Gaussian quadrature to evaluate

$$\int_{-1}^1 \frac{x^2}{1+x^4} dx \approx 0.487495494$$

Compare this with Simpson's 1/3-rule up to the same precision. [5]

3. Use appropriate Gaussian quadrature to evaluate the following integral with $n = 5$. [5]

$$\int_0^\infty \frac{e^{-x}}{1+x} dx, \quad \text{analytical value} \approx 0.59635$$

4. Integrate the following with appropriate n -point Gauss-Laguerre

$$\int_0^\infty e^{-x} x^4 dx$$

Compare with the analytical result $\int_0^\infty \exp(-x) x^n = n!$ [5]